

(TR-104) PYTHON WITH AI/ML

Training Day 89 Report

Date: 14 May 2026

Topic Covered: AI Attendance System Testing, Student Authentication Analysis, Project Enhancement Research & Data Analytics Curriculum Development

The eighty-ninth day of training focused on testing AI-based attendance workflows, reviewing teacher and student system functionalities, analyzing authentication mechanisms, exploring project enhancement opportunities, and organizing a Data Analytics curriculum structure.

- Performed testing of classroom attendance processing using uploaded student photographs.
- Validated face recognition accuracy during attendance scanning and verification processes.
- Tested student presence and absence marking workflows within the attendance system.
- Verified attendance submission, confirmation, and record-saving functionality.
- Examined teacher dashboard navigation and attendance management workflows.
- Reviewed subject-wise attendance tracking, attendance history, and record management features.
- Observed session state handling and workflow transitions during dashboard operations.
- Analyzed FaceID-based student authentication and recognition mechanisms.
- Verified student identification workflows using uploaded images and facial recognition techniques.
- Reviewed login redirection, access control, and authentication flow after successful recognition.
- Analyzed onboarding workflows for students whose face data was not available in the system.

- Studied attendance result dialogs and review mechanisms implemented in Streamlit.
- Verified attendance entry creation, database synchronization, and backend storage workflows.
- Reviewed attendance data processing and validation before final submission.
- Explored enhancement opportunities for existing MB AI Software project modules.
- Identified potential improvements for user interaction, workflow efficiency, scalability, and future development.
- Structured Data Analytics topics according to learning dependencies and concept progression.
- Reviewed Python, SQL, data visualization, and analytics-related concepts for curriculum enhancement.
- Organized learning modules to improve clarity, industry relevance, and practical implementation flow.
- Revised concepts related to AI attendance systems, face recognition workflows, authentication mechanisms, database synchronization, project enhancement strategies, and Data Analytics learning paths.

Overall, the training enhanced understanding of AI-based attendance management, facial recognition authentication, educational system workflows, curriculum design, and project improvement strategies.